

PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student with hands-on experience in Computer Vision, Embedded Systems, Industrial Automation, and Semiconductor Technologies. Interned at Semi-Conductor Laboratory (SCL), Mohali, where I developed a Vision-Based Orientation Detection System using OpenCV, Python, and STM32-based automation. Experienced in Docker, Linux environments, industrial control systems, and automation projects. Passionate about solving real-world engineering challenges through intelligent systems and emerging technologies.

EDUCATION

B.Tech | ECE | Kurukshetra University (CGPA: 6/10)
12TH | Science| Suraj School (83%)

2022 – 2026
2022

EXPERIENCE

- Project Intern @Semi-Conductor Laboratory (SCL), Mohali

Jan 2026 – May 2026

 - Developed a Vision-Based Orientation Detection System using OpenCV and Python.
 - Designed automation architecture using STM32 microcontrollers and motion-control components.
 - Implemented image-processing algorithms for component detection and orientation analysis.
 - Gained exposure to semiconductor manufacturing and industrial automation workflows.
- Onsite Engineer Intern @Echelon Edge Pvt. Ltd.

Jul 2025 – Aug 2025

 - Assisted in deployment, troubleshooting, and technical support activities.
 - Worked with Linux-based systems and Docker container environments.
 - Supported system maintenance and engineering operations.

PROJECTS

- Vision-Based Orientation Detection System

Python | OpenCV | STM32

 - Developed a system to detect and analyze the orientation of electronic components.
 - Integrated computer vision with automation concepts for industrial applications.
 - Designed a GUI for real-time visualization and monitoring.
- PLC-Based Pick and Place Machine

PLC | Sensors | Motor Drives | Automation

 - Developed and assembled an automated Pick and Place Machine during industrial training.
 - Worked with relays, motor drives, sensors, control panels, and industrial automation components.

LEADERSHIP & ACTIVITIES

- Design Lead @E-Cell UIET KUK

 - Led branding and design activities for workshops, events, and entrepreneurship initiatives.
 - Coordinated with multiple teams to ensure consistent communication and event promotion.
 - Managed creative projects and visual identity for student-led activities.

TECHNICAL SKILLS

Programming: Python Computer Vision: OpenCV, Image Processing Embedded Systems: STM32, Microcontrollers Automation: PLC, Industrial Control Systems, Motion Control	Tools & Platforms: LabView, Matlab, NI Test Stand Docker, Linux, Git Engineering: Robotics, Semiconductor Technology, Industrial Automation
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